



Fig. 5: ACS714 Hall effect-based linear current sensor. This simple carrier board for Allegro's $\pm 30A$ ACS714 Hall effect-based linear current sensor accepts a bidirectional current input with magnitude up to 30A. The output is an analogue voltage (86 mV/A) centered at 2.5V with a typical error of less than 1.5 per cent

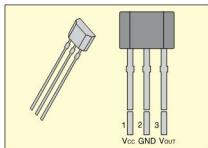


Fig. 6: A1301/A1302 linear Hall effect sensor IC by Allegro

switches on in the presence of a sufficiently strong south-pole magnetic field facing the marked side of the chip package. Similarly, the output switches off in the presence of a weaker south field and remains off with no field.

Unipolar types are optimised for applications of solid-state switches. Magnet proximity sensor for reed switch replacement in low-duty-cycle applications is a good example.

Transfer function

The transfer function of a device describes its output in terms of its input.

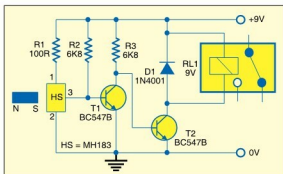


Fig. 7: Magnet-operated relay using MH183 unipolar Hall effect switch. The relay energises in the presence of a sufficiently strong south-pole magnetic field facing the marked side of the sensor MH183. Circuit is designed and tested by the author



Fig. 8: Open-loop Hall effect current sensor by Honeywell

For analogue-output Hall effect sensors, the transfer function expresses the relationship between a magnetic field input (in gauss; unit of magnetic flux density equal to 1 maxwell per square centimetre) and a voltage output.

The principal input/output characteristics of the digital-output Hall effect sensors are operate point, release point and the differential. As the magnetic field is increased, there is no change in the sensor output until the operate point is reached. Once the operate point is reached, the sensor changes

state to 'on'. Further increase in the magnetic input beyond the operate point will have no effect.

If the magnetic field is decreased below the operate point, the output remains the same until the release point is reached. At this point, the sensor's output returns to its original state (off).

The purpose of the differential between the operate and release points is to eliminate false triggering caused by minor variations in the input.

Practical applications

Hall effect sensors are widely used for speed, linear position, linear angle and position measurement in automotive, industrial and consumer applications. These devices can be used to sense many physical parameters—ranging from direct measurement of a magnetic field to detection of ocean currents. Hall effect sensors have been successfully used in linear-output applications such as current sensing (motor control protection and disk drives), position sensing (contactless potentiometers and brushless DC motors) and encoded switches (rotary encoders).

Digital-output applications include lens position sensors, proximity sensors, pressure sensors, limit switches, door interlocks and vending machines. ●

The author is a freelance writer and regular contributor to EFM



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